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Self-powered co-harvesting for batteryless sensing and monitoring in two-wheeled IoT applications



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G. Pasquini, M. Doglioni, A. Saviane and D. Brunelli (2025), "Self-Powered Co-Harvesting for Batteryless Sensing and Monitoring in Two-Wheeled IoT Applications," in IEEE STAR 2025.

- Good afternoon, and thank you for being here. I'm Giuseppe a master student at the university of Trento and I will present our work on **self-powered co-harvesting** for cycling applications.

Cycling is becoming an IoT platform

Electronics are now part of modern bikes

Sensors and
telemetry

Wireless controls
and actuators



This integration delivers tangible improvements across critical functional domains:

- Real-time **performance monitoring**
- Improved **safety enhancement**
- Smarter **bike setup optimization**



[BYB telemetry](#) & [Fox Transfer Neo Remote controller](#)

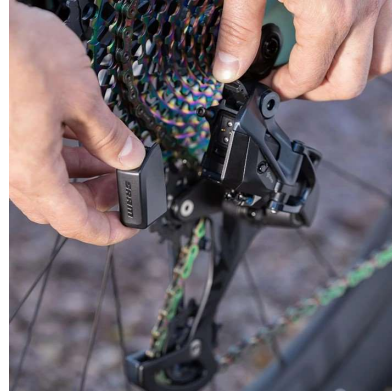
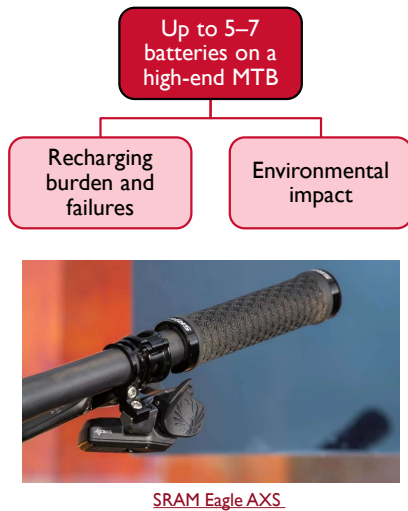


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- Modern bikes increasingly depend on electronics, but these systems rely on **multiple small batteries** that require frequent charging and maintenance. Our goal is to **eliminate this dependency** by harvesting energy directly from the bike **without affecting riding performance**.
- I want to highlight that we focus on **pedal-powered bikes**, not e-bikes, which already include a large energy source.
- Nowadays, cycling is rapidly evolving into an **IoT environment**, where sensors, wireless control units, and smart components improve **performance, safety, and bike setup optimization**.
- These technologies deliver measurable benefits to riders and teams. However, this also means more devices to power, more electronics, and **greater energy demand** on the bike.
- This emerging scenario **highlights a key challenge**: powering these connected systems **without adding more batteries**.

The problem: too many batteries



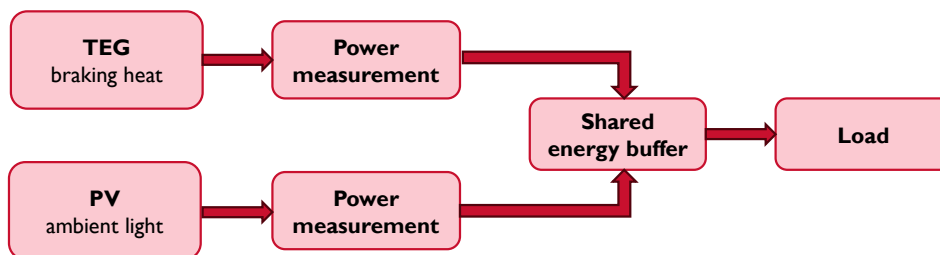
We need autonomy, not more batteries



- Today, a high-end mountain bike and e-bikes may host **five to seven batteries**—for sensors, controllers, suspension systems, and wireless accessories. This leads to **charging burden**, failures during rides, and unnecessary **environmental impact**.
- The rider must remember to charge everything, and even a single dead battery can disable a critical function.
- Instead of increasing battery capacity, we believe the right direction is **autonomy**: In particular, powering onboard electronics through **locally harvested energy**, directly from the physical phenomena of the ride.

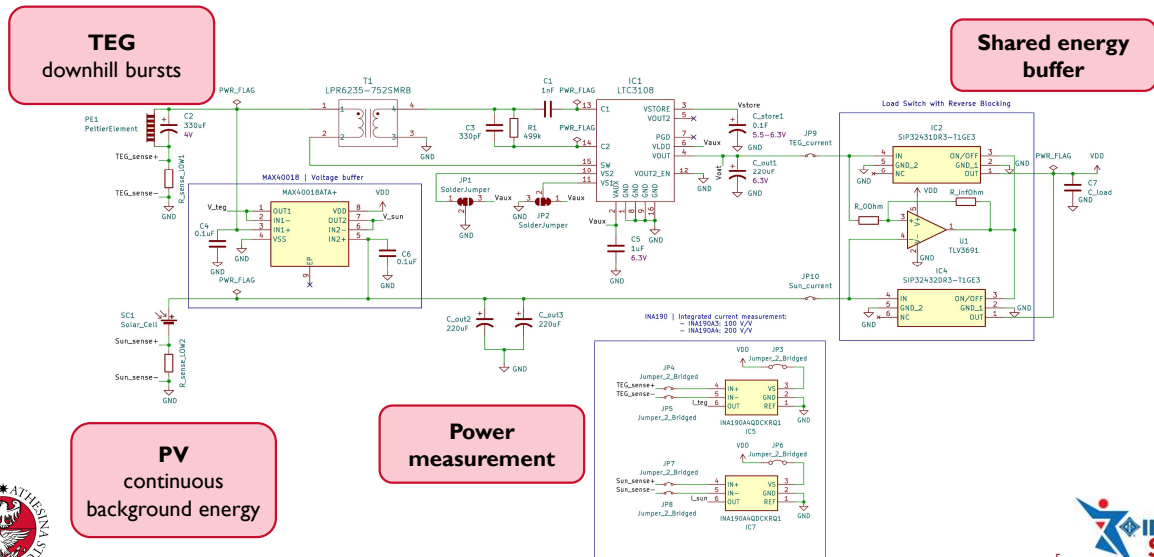
Core idea

- Hybrid harvesting using **TEG** on brake pads + small **PV cells**
- The two sources are **naturally complementary** in mountain biking
- Provides **friction-free energy** under highly variable conditions
- Harvesting adapts to both braking events (TEG) and ambient light (PV)
- **Power sources also act as sensors** (TEG $\leftrightarrow$ brake temperature $\leftrightarrow$ braking events)



- Our core idea is a **hybrid co-harvesting architecture** that combines **thermoelectric generators** on brake calipers with **small photovoltaic cells**.
- These two sources are **naturally complementary** in mountain biking: braking events produce thermal bursts for the TEG, while sunlight provides a continuous background through PV.
- This enables **friction-free energy harvesting** in highly variable outdoor conditions. Additionally, these power sources can act as **indirect sensors**, since TEG output correlates with **braking behavior**.

Hybrid system concept



- The hybrid approach works because each source compensates for the other.
- **Downhill riding** produces short **mW-level bursts** through braking heat, while during **climbs or flat sections**, sunlight ensures **baseline energy** from the PV.
- A **shared energy buffer** and power management block stabilize the output, ensuring a **predictable and usable power profile**. This keeps the system powered for sensing and wireless data, even when the harvesters produce non constant power.

Solar harvesting (PV)

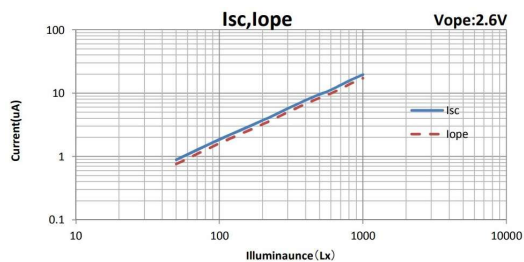
- Specific **PV** for **low-light conditions**, including cloudy or indoor environments
- Generates μW to mW depending on sunlight
- Provides **baseline power** over the full ride



PV (AM-1606C-MEL), mounted on BFTAG01.A sensor tag based on STM32L0

Output power across common lighting conditions:

- **Indoor/shade:** $\sim 8\text{--}30\ \mu\text{W}$
- **Daylight:** $\sim 170\ \mu\text{W}$
- **Direct sunlight:** $\sim 1\ \text{mW}$



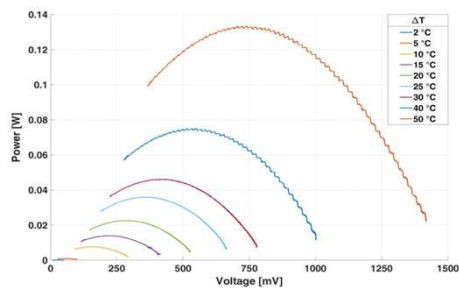
PV Current vs. Illuminance at Vope



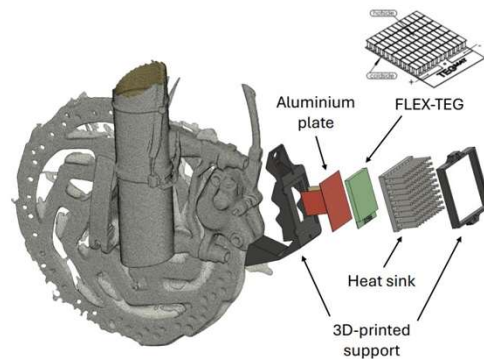
- We choose Specific **PV** for **low-light conditions**, that operates across a wide range of illumination—from **low-light** to **full sunlight**, generating from **μW up to the mW range**.
- PV ensures **background power** throughout the ride, keeping the node alive even when the rider is not braking.
- This complementarity is important for a **reliable energy availability** in our architecture.

Thermoelectric harvesting (TEG)

- Mounted on the brake caliper using the natural **temperature gradient (ΔT)**
- Generates **mW-level bursts** during braking events
- Output directly follows **thermal dynamics**



Lab characterization, power generation depending on ΔT .



TEG supporting structure



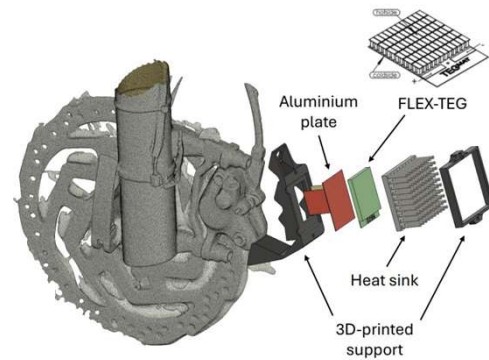
- Braking converts **kinetic energy into heat**, and this heat is normally wasted. By placing a TEG across a temperature gradient, we can **convert thermal variations into electrical power**.
- In the lab characterization, we measure the **output power over different temperature gradient**, and we observed **up to tens or over 100 mW** depending on the temperature difference.
- These bursts are ideal for **sensing and BLE transmission**, especially during downhill segments, where data acquisition and system reactivity are most important.

Real TEG setup on the bike

- Custom **3D-printed mount**
- **Aluminum plate** in direct contact with the brake caliper (hot side)
- **Heat sink on the cold** side to maximize ΔT and harvesting efficiency
- TEG mounted **without friction** on the wheel system



TEG setup



TEG supporting structure



- To validate this concept, we built a **custom 3D-printed mount**.
- An **aluminum plate** provides optimal thermal contact on the hot side, while a **heat sink** boosts the temperature difference on the cold side. The structure avoids any **friction with moving parts**, enabling **stable on-field thermal harvesting**.
- This ensures the TEG can operate effectively during real downhill mountain biking conditions.

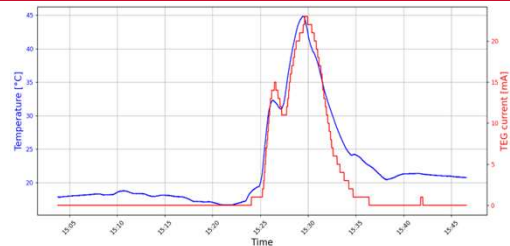
TEG on-field results

Ride 1

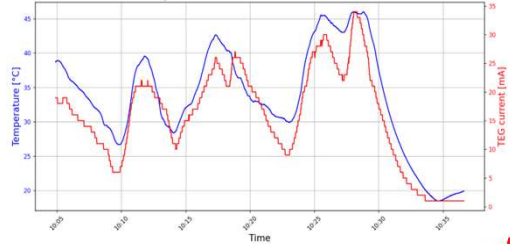
- 1.5 km downhill segment
- Maximum TEG hot-side temperature: 45°C
- Ambient temperature: ~20°C
- **Peak power: 2.3 mW**
- **Harvested energy: 439 J**

Ride 2

- longer descent (~6 km),
- Maximum TEG hot-side temperature: 46°C
- Ambient temperature: ~16°C
- **Peak power: 3.2 mW**
- **Harvested energy: 2236 J**



Hot side temperature and current production over time in RIDE 1



Hot side temperature and current production over time in RIDE 2



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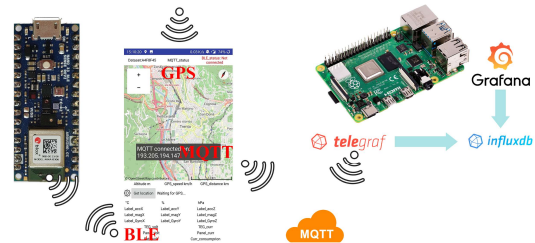


- We tested the system on real downhill routes with different lengths and ambient conditions.
- A **strong correlation** was observed between the **hot-side temperature** (*blue line*) and the **TEG current** (*red line*) — confirming the link between **braking activity** and **energy generation**.
- In both cases, the **TEG delivered peak power between 2 and 3 mW**, with a **significant amount of harvested energy**.
- The **real-world performance is lower than laboratory potential**, mainly because of the **first, non-optimized thermal setup**.
- Nevertheless, the harvester already provides **sufficient energy for batteryless MTB telemetry**, validating the concept **under real riding conditions**.

What we do with the energy: Two Use Cases

Use Case #1: IoT pipeline for batteryless telemetry

- **Bike setup** & configuration
- **Performance** monitoring
- Track **real-time data** on usage
- **Data visualization** & insights



BLE → Smartphone App → Cloud (MQTT) → Dashboard for Real-time data and visualization

Use Case #2: Power commercial actuators and remote controllers

- Electronic shifting systems
- Seatpost droppers



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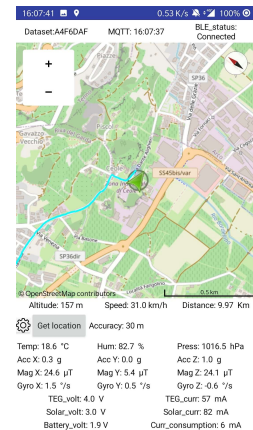
- After validating the harvesting system, we focused on **how to use the collected energy effectively**.
- To **demonstrate its practical value**, we validated **two main use cases**.
- **First**, a **complete IoT telemetry pipeline** — including **sensing**, **BLE transmission**, **smartphone app**, **cloud upload**, and **dashboard visualization**.
- **Second**, powering **commercial on-bike components** such as **wireless controllers**, **Seatpost droppers** and **shifters**, proving that the harvested energy can support **real, rider-visible functions**, not just sensor data.

Use Case #1: Batteryless Telemetry

Harvested energy powers a self-sustaining sensor node for sensing, processing and wireless transmission.

Strategy: Energy is buffered on a capacitor and the node wakes only when sufficiently charged

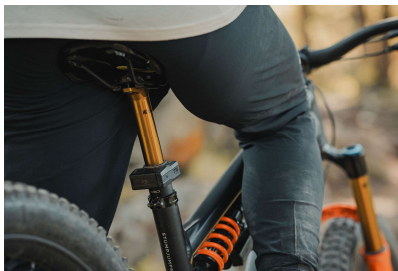
- **Required energy per cycle: 0.133 mJ** Brake temperature, IMU, suspension
- Continuous sampling during downhill (TEG with $\Delta T > 20^\circ\text{C}$)
- 0.77 s sampling period with PV only (10.000 lux)



- In this first use case, the **harvested energy powers a fully self-sustaining sensor node**, handling **sensing, processing, and wireless transmission** without any battery.
- The strategy is simple: the energy is **stored in a capacitor**, and the node **wakes up only when enough charge is available**. The system needs just **0.133 mJ per transmission cycle**, which allows us to monitor **brake temperature, IMU data, and suspension behavior**.
- During downhill sections, when **ΔT exceeds 20°C** , the **TEG enables continuous sampling**, while under daylight conditions, **PV alone still supports a sub-second sampling**.
- These results demonstrate **true energy-neutral, batteryless telemetry** in real riding scenarios.

Use Case #2: Powering Commercial Devices

- **Goal:** Use harvested energy to drive commercial on-board actuators and wireless components (today battery-powered)
- **Works today** (low-power): remotes, shifters, power meters (**1–4 mW**)
- **Not yet** (high-power): droppers, derailleurs, active suspension (**30–300 mW**)
- Device optimization is key: a dropper can work battery-free with short, limited pushes powered by a capacitor; but the trade-off is reduced continuous actuation.
- Continuous actuation: needs optimization.



Fox Transfer Neo Dropper



SRAM Eagle AXS



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- In this second use case, we can already power **low-power commercial systems** such as remotes and power meters, typically in the **1–4 mW range**.
- While, Higher-power actuators, like **droppers or shifters**, require **up to hundreds of milliwatts**, which is still challenging. However, using **short capacitor-based pushes**, we can enable limited actuation even without a battery.
- With further consumption optimization, we expect to **close this gap** between harvesting capability and actuation demand.

Key takeaways

Hybrid harvesting

stable & reliable power profile

Results

zero friction, zero maintenance, zero charging

Telemetry

batteryless today (fully supported)

Commercial devices

partially supported (optimization needed)



- To sum up, our work addresses the growing **battery problem** in connected bikes by introducing **a hybrid co-harvesting architecture** that provides a stable and reliable power supply, **eliminating charging** and **minimizing maintenance**.
- Today, **batteryless telemetry is fully supported**, while **commercial actuation is only partially supported and still evolving**.
- Most importantly, **we validated the entire system outdoors, on real bikes and not just in theory or in the lab**.

Conclusions & future work

- **Improve heat transfer and TEG efficiency** to approach lab-scale performance in real riding conditions
- Add more energy sources like **vibration energy harvesting** for rough terrains in MTB use cases
- Extend the architecture to e-bikes and **additional sensors** for telemetry
- Move toward fully batteryless on-bike electronics for maintenance-free IoT systems



- In conclusion, Our next steps focus on **improving thermal transfer** to bring on-bike performance closer to lab characterization, and on **adding new energy sources**, such as **vibration harvesting**.
- We will also **extend the architecture** to e-bikes and additional sensors.
- The long-term vision is a **fully batteryless electronic system**, where bicycle IoT devices are **maintenance-free**, powered entirely by the physics of the ride **without affecting riding performance**.



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Thank you for your attention

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